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Quantitative modeling of scratch-induced deformation in amorphous polymers

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ABSTRACT

In order to predict scratch performance of polymers, the present study focuses on quantitative assessment of various scratch-induced deformation mechanisms based on a set of model amorphous polymers *via* numerical modeling. A modification of Ree-Eyring theory is used to account for the rate dependent behavior of the model polymers at high strain rates using the experimental data obtained at low strain rates. By incorporating the rate and pressure dependent constitutive and frictional behaviors in the finite element methods (FEM) model, good agreement has been found between FEM simulation and experimental observations. The results suggest that, by including appropriate constitutive relationship and frictional model in the numerical analysis, the scratch behavior of polymers can be quantitatively predicted with reasonable success. Usefulness of the present numerical modeling for designing scratch resistant polymers is discussed.

1. Introduction

Scratch-induced surface deformation in polymeric materials is a complex mechanical process due to their strong rate, temperature and pressure dependent responses. The mechanical behavior of polymers generally exhibits strain softening-strain hardening phenomena, different constitutive behaviors under tension and compression, rate, temperature and pressure dependent yielding, and viscoelasticity. Surface friction also greatly influences the scratch-induced deformation features as changes in coefficient of surface friction alter the stress state polymer substrate experiences near the surface during the scratching process. Since the development of scratch-induced damage features in polymers involves large-scale deformation, close-form solutions are inadequate for modeling scratch behavior of polymers. Consequently, extensive research work has been carried out to study the evolution of scratch-induced deformation features and learn how they are affected by the bulk mechanical and surface properties [1–13].

Jiang et al. [1] showed that the development of different scratch-induced deformation features depends on the polymer type. The stress analysis using finite element methods (FEM) showed that the same material point could successively undergo tension and compression as the scratch tip passed through it. Based on their experimental observation, a polymer scratch damage evolution map was developed to qualitatively differentiate the scratch behaviors of polymers with variation in material constitutive relations. Similar scratch deformation maps, developed by Briscoe et al. [14,15] using conical indenter and constant/dead weight scratch normal load, showed evolution of different scratch-induced deformation to vary with the scratch speed due to changes in imposed strain rate. Using FEM parametric studies along with the ASTM scratch test [16] on a set of model polymers, Hossain et al. [4,5,8] showed that yield stress, strain at stress recovery and strain hardening slope beyond the strain at stress recovery in compression are the most important parameters that influence the residual scratch depth and shoulder height development along the scratch path. Tensile behavior has little influence on residual scratch depth and shoulder height formation but affects the surface roughness on the scratch groove along the scratch path, which was primarily caused by micro-cracking in the polymers investigated. Poisson's ratio and Young's modulus, in the range of 1.65 GPa–4 GPa, have shown not to significantly influence the residual scratch depth using FEM modeling [3]. Bucaille et al. [6] employed experimental work and FEM to conclude that a larger strain hardening led to greater elastic deformation, thus less

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cumulative plastic strain [6,17]. Bellemare et al. [18] reported a decrease in pile-up height (shoulder height) with the increase in strain hardening exponent using pure Copper and Copper/Brass alloy. Browning et al. [2] showed that, using experimental study on styrene-acrylonitrile (SAN) random copolymers, an increase in acrylonitrile (AN) content or molecular weight (MW) can have a positive effect on improving the scratch resistance as it increases the tensile strength and ductility.

Using their analytical expressions for stress field due to a circular contact region carrying a hemispherical Hertzian normal pressure and a proportionally distributed shearing traction, Hamilton and Goodman [19,20] showed that an increase in surface friction intensifies and moves the maximum stresses from sub-surface toward the surface, and, thus, inducing greater deformation. According to the study, a maximum tensile stress also develops at the rear end of the circular contact when increasing the surface friction, which can be thought of responsible for the ring crack observed in brittle materials. An extensive study [7] on the effect of surface friction on scratch depth and shoulder height development has shown that the coefficient of adhesive friction indeed affects the onset and extent of scratch depth and shoulder height, but the degree of influence can be altered by modifying the constitutive relation.

Most, if not all, of the experimental studies carried out by the researchers to correlate mechanical properties with the evolution of various scratch-induced deformation features can be considered qualitative since the bulk mechanical properties used to draw the conclusion were measured at a strain rate much lower than the rate polymer surface would experience at the imposed scratching speeds. Similar conclusion can be drawn in the case of FEM simulation efforts as most of the studies employed simplistic constitutive model in their analyses which did not take into account the rate, temperature and pressure dependent behaviors of polymers. Furthermore, simplistic description of the contact between the tip and the substrate was employed using the Coulomb's law of friction. As a result, most of the numerical analyses emphasize only on qualitative comparison between the FEM simulation and experimental findings.

To quantitatively predict the scratch behavior of polymers using FEM, strain softening-strain hardening phenomena, asymmetric behavior in tension and compression, rate, temperature and pressure dependent behaviors, viscoelasticity, to name a few, will have to be incorporated in the constitutive relations. The contact between the tip and the polymer substrate has to be modeled properly to account for not only the coefficient of friction due to adhesion but also the friction developed due to large-scale material deformation. The scratch tip geometry and surface roughness of the polymer substrate also plays an important role during the contact between the rigid tip and polymer substrate. A quantitative model for predicting scratch behavior of polymers can then be developed if all these features can be included in the FEM simulation.

The objective of this study is to quantitatively predict the scratch-induced deformation of amorphous polymers *via* FEM by including the key characteristics of polymer constitutive behavior and an appropriate contact mechanics model. It is hoped that the present findings can facilitate development of scratch resistant polymers.

2. Theoretical consideration

During the scratching process, the effective strain rate on or near the surface can be approximated by Ref. [14]:

$$\dot{\varepsilon} = \frac{v}{w} \tag{1}$$

Where, $\dot{\varepsilon}$ is the effective strain rate, v is the scratch speed and w is the scratch width. For a scratch speed of 100 mm/s and spherical scratch tip of 1 mm diameter, as recommended by the ASTM standard for automotive applications [16], the effective strain rate on or near the surface can reach at 100s to 1000s 1/s. Thus, to quantitatively predict the scratch-induced deformation of polymers using FEM, high strain rate constitutive behavior of polymers is needed. It should be noted that, since the strain rate varies from one position to another during the scratching process, the mechanical behavior only at a fixed strain rate is inadequate for modeling the scratch behavior. The constitutive behavior of the polymer substrate from low to high strain rate is needed along with an interpolation scheme to account for the wide strain rate range the material experiences.

For amorphous polymers, the yield stress generally increases with strain rate and a linear relationship can be established between the yield stress and logarithm of strain rate. However, the post-yield behavior can remain unchanged with the increase in strain rates [21–25]. To describe the rate dependent yield or flow stress for glassy, amorphous polymers, several theoretical models are available. Although based on specific molecular mechanisms involved in the yield behavior of polymers, Robertson [26] and Argon [27] models found only limited success in describing the rate dependent plasticity in amorphous polymers. Perhaps the most widely-accepted model to describe the rate dependent plasticity in amorphous polymers is the Ree-Eyring model [28], a modification of the original Eyring model [29]. The ability to change the slope at an intermediate strain rate distinguishes the Ree-Eyring model from Robertson and Argon models, and, thus, allows better prediction of the rate dependent plasticity in amorphous polymers from moderate to high strain rates. According to the formalism of the Ree-Eyring theory, yielding of amorphous polymers can be considered dependent on particular degrees of freedom in polymer chains, whose relaxation or activation depends on the applied strain rate and temperature. At low strain rates and/or high temperatures, pure viscous flow can be considered to take place at yielding due to jumps of segments of the macromolecule from one equilibrium position to another, called the α process. For high strain rates and/or low temperatures, the yield behavior of an amorphous polymer can be assumed of involving two rate processes, α and β, and the plastic flow due to these processes can be considered additive. At high strain rates and/or low temperatures, pure viscous flow also takes place at the yielding due to jumps of segments of backbone chain of the macromolecule from one equilibrium position to another similar to the single activated α process. But, the α process is hindered as the molecular movements are partially frozen even when the stress around the yield stress is applied. To activate these movements, it is necessary to supply additional energy by applying additional stress. Thus, at high strain rates and/or low temperatures, the observed yield stress can be considered as the sum of two stresses with respect to the α and β processes.

Using the commercially available polycarbonate (PC) samples (Makrolon from Bayer), Bauwens-Crowet et al. [30] described the yield behavior of amorphous polymers by assuming the involvement of two different flow processes. They found success when compared their experimental data by curve-fitting using the Ree-Eyring model for multiple rate activated process. It was suggested [30] that the β flow process observed at yielding is the same process observed in damping tests and can be correlated with the dynamic mechanical loss peak, i.e., loss tangent *vs.* temperature at a given frequency. They reported that the value of the activation energy Q_β for PC was in agreement with the value reported from dielectric measurements [30]. They found similar results for PVC, as well [31]. According to Bauwens et al. [32], unlike Q_β, Q_α may not be compared with the activation energy related to the primary transition observed in dielectric or mechanical damping tests. Using PC, Bauwens [33] showed that the β yield process and β peak revealed by the oscillatory damping tests are from the same molecular movements, and, thus, the activation energies associated with the respective processes should be equal. Similar yielding behavior has been described by the Ree-Eyring model for PC, PVC, PEMA and PMMA [31,34–37]. Experimentally, similar yielding behavior based on the transition of slope depending on the strain rate at a particular temperature that shows two distinct slopes has been reported in literature for PC [21,22,24,25,38], PMMA [21,38], PP [39,40], TPO [41], PAI [38] and PVDF [22]. Similar description has also been reported for Young's modulus [21,41,42].

Mulliken and Boyce [25] performed dynamic mechanical analysis (DMA) on PC samples to correlate the strain rate dependent β transition peak with the transitional strain rate where the single activated process in amorphous polymers changed into two rate processes. Analytical expressions for the elastic moduli of PC were developed by first decomposing the storage modulus reference curves obtained from DMA into their respective α and β components. As the curve was traced with decreasing temperature, a significant upturn was observed, which can be correlated with the onset of restriction of the β-process. Using this analysis, the PC storage modulus reference curve was divided into α and β components. These α and β components were then considered to shift with strain rates by the amounts determined using the shift calculated in the respective transition peak. The entire modulus curve was then reconstructed for any strain rate by first shifting the components of the reference curve by the appropriate amounts, and then adding the components. This decompose/shift/reconstruct (DSR) method enabled extrapolation of the elastic modulus of PC at temperatures and strain rates well beyond the capabilities of the DMA instrument. By assuming that the significant material transition is due to the restriction of the same molecular mechanism associated with the β-transition, the shift in β-transition can be used to predict the strain rate at which the yield behavior of amorphous polymer undergoes a transition. The transition strain rate predicted by the DSR method was consistent with the compressive yield stress data obtained through experimentation *via* Instron and split-Hopkinson bar tests [25].

Since the stress state during scratch is generally multiaxial, the pressure dependent mechanical behavior of polymers and volume change beyond yielding is another area of concern when modeling polymer scratch. Spitzig and Richmond [43] studied the stress–-strain response of high density polyethylene (HDPE) and PC in tension and compression, and found that the strength differential between tension and compression was due to the pressure dependent yielding. They suggested that the flow stress is linearly dependent on mean pressure, similar to that of the Young's modulus. They also observed a negligible inelastic volume change, similar to the findings reported by others [44,45].

Several theoretical models have been developed to capture the rate dependent stress–strain relationship of glassy polymers with limited success [25,46–48]. To the best of our knowledge, the mathematical models developed for glassy polymers are unable to simultaneously describe the rate and pressure dependent stress–strain response. Also, these models usually employed curve-fitting based on experimental data to determine the required material constants, which makes these approaches cumbersome.

Since scratch-induced deformation involves sliding indentation of a rigid asperity over the surface of polymers, the contact between the tip and the substrate, the surface roughness of the contacting bodies, the evolution of friction and the dissipation of frictional energy need careful consideration for quantitative modeling *via* FEM. As the tip interacts not only with the surface of the substrate but also with the sub-surface, the Coulomb's law of friction may be insufficient to describe the frictional behavior involved in the scratching process. Bowden and Tabor [49] first proposed, based on the assumption that there is negligible interaction between adhesion and deformation process during sliding, that the total intrinsic coefficient of friction (μ_i) can be considered as the summation of friction due to adhesion (μ_a) and friction due to deformation (μ_d) [50].

$$\mu_i = \mu_a + \mu_d \tag{2}$$

For ductile materials, the coefficient of friction due to adhesion (μ_a) for dry contact can be written as [49]:

$$\mu_a = \frac{A_r \tau_a}{F_n} = \frac{\tau_a}{P_r} \tag{3}$$

Where, A_r is the real area of contact, τ_a is the shear strength and P_r is the mean real pressure. Again, for polymers, due to the rate dependent response, τ_a is dependent on strain rate. However, it has been shown that, τ_a generally remains almost unchanged or, in some cases, decreases at room temperature with an increase in sliding velocity [51,52]. For polymers, the shear strength, τ_a can be considered [52] as a linear function of mean contact pressure, P_r:

$$\tau_a = \tau_0 + \gamma P_r \tag{4}$$

From Equation (3),

$$\mu_a = \frac{\tau_0}{P_r} + \gamma \tag{5}$$

Where, τ_0 is the intrinsic characteristic shear strength and γ is the pressure coefficient. Bowers [53] showed that the values of coefficient of friction as a function of pressure in sliding contact can be determined from shear strength as a function of pressure using friction measurement of thin polymeric films and bulk shear data. Briscoe and Tabor [54] studied the effect of pressure on the shear properties of thin polymeric films (HDPE and PMMA) by conducting frictional experiments with varying tip geometry and contact loading. They found that increase in pressure increases the shear yield stress linearly following Equation (4). They compared their data with those obtained for torsional shear of bulk polymers under various hydrostatic pressures. They found that the value of pressure coefficient, γ, remains almost same in both type of experiments. Similar findings was also reported in literature [51,55]. Duckett et al. [56] studied the torsional behavior of PC under superposed hydrostatic pressures and found that the shear yield stress increases linearly with pressure. Using the torsion tests, similar behavior has also been observed in PMMA [57]. For ductile materials, the coefficient of friction due to deformation (μ_d) for a spherical rigid asperity of radius R in contact with a softer substrate can be written as [52]:

$$\mu_d = \frac{4}{3\pi}\frac{a}{R} \tag{6}$$

Where, a is the contact radius. For a relatively large width of the groove compared to radius of sphere [58]:

$$\mu_d = \frac{2}{\pi}\left\{\left(\frac{R}{a}\right)^2 \mathrm{Sin}^{-1}\left(\frac{a}{R}\right) - \left[\left(\frac{R}{a}\right)^2 - 1\right]^{1/2}\right\} \tag{7}$$

As can be deduced from the equations, μ_d increases as the tip goes deeper. Although the material piles up in front of the tip in many cases during the scratching process, the model of rigid spherical asperity to calculate μ_d neglected the pile-up of material in front of the tip. However, the contribution from material pile-up in front of the tip can be significant in some cases, specifically, at high normal load.

The effect of chain orientation on the frictional development during sliding contact also has to be considered in case of polymer scratch. Tabor and Williams [59] studied the evolution of friction during sliding of the PTFE samples using hemispherical slider. They concluded that, since the shear strength across the chains was higher than the shear strength along the chains, the friction is higher across the molecular chains. They also found that the deformation component of friction is independent of the direction of sliding. Bely et al. [60] showed that the frictional value is higher in the unoriented polymer compared to the oriented one. Since the coefficient of friction due to adhesion can be correlated with the bulk shear strength, as described earlier, the effect of chain orientation in polymers can be included by taking into account orientation dependent shear strength.

Surface roughness of the contacting bodies is also important, specifically in multi-asperity contacts as it determines the real area of contact, and, thus, evolution of friction. But, for scratching using 1 mm diameter spherical tip, the normal load required to induce plastic contact is comparatively low. For plastic contacts, the coefficient of friction due to adhesion can be considered independent of surface roughness [52].

In any sliding interaction, most of the input frictional energy is generally consumed during plastic deformation which is then converted to heat in materials close to the interface [52]. In dry contact, this heat is conducted into two sliding members through contact spots with a high temperature gradient. Since the mechanical and frictional properties of polymers depend on operating temperature, the temperature rise due to frictional heat dissipation is also important. As the temperature rise at the contacts is transient and difficult to measure accurately, a number of theoretical models have been developed [49,61–65] to predict the temperature at the asperity contacts. It has been shown that, in the case of polymers, the temperature rise is small due to reduction in heat generation per unit area as a consequence of relatively large area of contact [66,67]. Using a polyetheretherketone (PEEK) ball of 19 mm diameter sliding on Sapphire disc, Wong et al. [68] recently measured the maximum temperature rise at the contact by employing a Infrared camera. They found negligible increase in temperature for 40 N load at a speed of 100 mm/s which is also the scratching speed carried out here. An increase of ~7 °C from room temperature was reported up to a speed of ~1 m/s at 40 N of normal load.

3. Experimental procedure

3.1. Model systems

SAN, in the form of reactor-grade random copolymers polymerized by free-radical reactions, and PC (Makrolon 2800 from Bayer MaterialScience) systems were provided by BASF SE (Ludwigshafen, Germany). The SAN model system contained 19 percent AN by weight. Molecular weight information of the model systems was provided by BASF SE and is summarized in Table 1.

The samples with dimensions of 150 mm × 150 mm × 6 mm and 150 mm × 150 mm × 3 mm were injection-molded plaques provided by BASF SE. Upon receipt, all the plaques were first dried in an oven in between two smooth glass plates at 80 °C for 6 h in vacuum. Then the samples were annealed at ~10 °C above their respective glass transition temperature (T_g) for 1 h in vacuum (T_g for PC is ~148 °C and for SAN is ~120 °C). Finally, the samples were slowly cooled to room temperature at 1.8 °C/min. This heat treatment procedure minimizes any residual surface stresses and eliminates any chain orientation induced during the injection molding process. The surface finish of the plaques was smooth with RMS roughness values of 26 nm and 45 nm for PC and SAN samples, respectively. The roughness value was measured on an area of 525 μm × 700 μm.

Table 1
Molecular weight information of the model polymers.

	SAN	PC
Acrylontrile content (wt%)	19	–
Weight average molecular weight, M_W (kg/mol)	134	67
Polydispersity	4.1	2.6

3.2. Mechanical property characterization

Uniaxial tension and compression tests were performed following the ASTM D638 [69] and D695 [70] standard, respectively. A screw-driven MTS® Insight load frame equipped with a 30 kN capacity load cell was used for all tests. MTS® Testworks 4 was used as the software interface for data collection.

For uniaxial tensile testing, plaques with dimensions of 150 mm × 150 mm × 3 mm were cut to dog-bone shape specimens with a nominal thickness of 3 mm and width of 6 mm. The samples were polished to make sure that no surface defects were present. Actual dimensions were measured with a digital micrometer caliper. The tensile testing was conducted at four nominal strain rates ~1.6 × 10^{-3}/s, 3.3 × 10^{-3}/s, 1.6 × 10^{-2}/s and 1.6 × 10^{-1}/s. An MTS® extensometer with a gauge length of 25.4 mm was used to monitor the displacement for strain calculations. At least three samples were tested at each strain rate.

Prismatic uniaxial compression specimens (12.7 mm × 6 mm × 6 mm) were prepared by precision-cutting using a diamond saw. After cutting the samples, the surfaces were polished using P2400 first, and, then P4000 grit silicone-carbide abrasive paper. Care was taken to ensure that all the edges were flat and square. The compression test was conducted at three nominal strain rates ~3.1 × 10^{-4}/s, 3.1 × 10^{-3}/s and 3.1 × 10^{-2}/s. An MTS® laser extensometer was used to monitor the displacement and strain. White lithium grease was used to provide sufficient lubrication to minimize contact friction between the fixture and the sample surfaces under compression. At least three samples were tested at each strain rate.

3.3. Dynamic mechanical analysis

Dynamic mechanical analysis (DMA) was performed on a TA Instruments ARES G2 Rheometer using a torsional fixture. Samples with nominal dimensions of 30 mm × 9 mm × 3 mm were used for the DMA tests. The specimens with a pre-tension of ~200 gm and a strain level of 0.05% were used. The samples were tested from −140 °C to 200 °C for PC and −140 °C to 160 °C for SAN. The samples were tested at frequencies of 0.1 Hz, 0.5 Hz, 1 Hz, 5 Hz and 10 Hz which correspond to strain rates of 2 × 10^{-4}/s, 1 × 10^{-3}/s, 2 × 10^{-3}/s, 1 × 10^{-2}/s, 2 × 10^{-2}/s, respectively [25]. Storage modulus and loss modulus information was recorded during the test and corresponding loss tangent value was calculated.

3.4. Measurement of surface friction coefficient

To determine the coefficient of surface friction, μ at the interface between the model systems and scratch tip, a flat smooth stainless steel tip with 10 mm × 10 mm square area was employed. The RMS roughness value of the stainless steel flat tip was 225 nm on an area of 525 μm × 700 μm. The flat tip was installed on the scratch machine and tests were conducted under 5 N constant normal load for a distance of 40 mm at a velocity of 100 mm/s. The flat tip was self-aligned during the testing. Five tests were conducted for each model polymers to obtain an average value of μ. This procedure to measure the coefficient of surface friction is comparable with the method described in literature [71].

3.5. Scratch-induced deformation

3.5.1. Scratch test

Scratch tests were carried out according to the ASTM D7027-05/ ISO 19252:08 standard [16] by employing a linearly progressive normal load of 1–70 N. A constant scratch speed of 100 mm/s was used for the scratch tests with a scratch length of 100 mm. A stainless steel spherical scratch tip of 1 mm diameter was used to conduct the tests. Five scratch tests were performed on the same plaque.

3.5.2. Microscopic observation

Immediately after completing the scratch tests, all the samples were scanned using a Keyence® VK9700 violet laser scanning confocal microscope (VLSCM) for high-resolution analysis of the scratch-induced damage mechanisms. The microscope is equipped with a 408 nm wavelength violet laser and has a height resolution of ~1 nm. The VK Analyzer software provided with the microscope was used to obtain optical images as well as topographical profiles. The scratch depth, shoulder height and scratch width at different locations on the scratch path were measured using the VK Analyzer software. The samples were scanned again after full relaxation (~8 months after the scratch test) to measure the percent relaxation or viscoelastic recovery.

The measurement of scratch-induced deformation was done by scanning at specific points along the scratch path using 20× magnification, which corresponds to an area of ~525 μm × 700 μm. Since the scratch-induced deformation also varies in this area, three measurements at the beginning, in the middle and at the end have been conducted to give an average value and standard deviation (vertical error bar in the figures) of scratch-induced deformation corresponding to a point. Although linearly increasing with scratch length, the scratch normal load has some fluctuation at a particular point. These points are averaged out to give an average value and standard deviation (horizontal error bar in the figures) of normal load corresponding to a point along the scratch path.

4. Experimental findings

4.1. Uniaxial tensile and compressive properties

Figs. 1 and 2 show the representative plot of compressive true stress–strain behavior and true yield stress of PC and SAN as a function of strain rate, respectively. The engineering stress–strain data obtained during the test was converted to true stress–strain by considering volume preservation during plastic deformation. As shown in Figs. 1b and 2b, compressive yield stress increases with strain rate for both systems. Also, compressive post-yield behavior remains almost unchanged with increase in strain rate (Figs. 1a and 2a). An increase in compressive modulus and yield strain in compression is observed for both systems with increase in strain rate [47]. Although not shown here, an increase in tensile yield stress is observed, whereas modulus and yield strain in tension remains essentially the same for PC [47]. SAN shows brittle behavior in tension, and, consequently, only tensile strength values can be obtained during the test. The tensile strength remains almost constant despite the increase in strain rate whereas the tensile modulus shows an increase with strain rate for SAN [47].

4.2. Dynamic mechanical analysis

4.2.1. PC system

Representative storage moduli and loss tangent plots in logarithmic scale obtained in DMA test for PC are shown in Fig. 3. Distinctive α and β transition peaks can be observed at ~150 °C and ~−96 °C at 1 Hz, respectively. The shape of the curves and the location of the transitions matches well with the data provided in the literature for PC [33,72]. Fig. 4 shows the plot of loss tangent as a function of temperature near the β transition zone for the five frequencies tested. The β transition shifted to higher temperature with increase in frequency. Similar behavior was also observed for α transition. Using the shift in β transition, the activation energy for β transition is calculated to be 9.31 kcal/mol, which matches well with the activation energy parameter used for modeling strain rate dependent yield behavior of PC by Bauwens-Crowet et al. [30]. As discussed by Bauwens [33] using PC (Makrolon from Bayer), the β yield process and β transition observed in DMA test are due to the same molecular movements. Thus, the respective activation energies associated with the processes should be equal, which is shown to be the case.

Fig. 5 shows the dependence of storage modulus on strain rate as a function of temperature in PC obtained *via* DMA test. As shown in the figure, similar to loss tangent plots in Fig. 4, the α and β transition temperatures increase with increase in strain rate. To find the onset of β relaxation, not the peak, the storage modulus

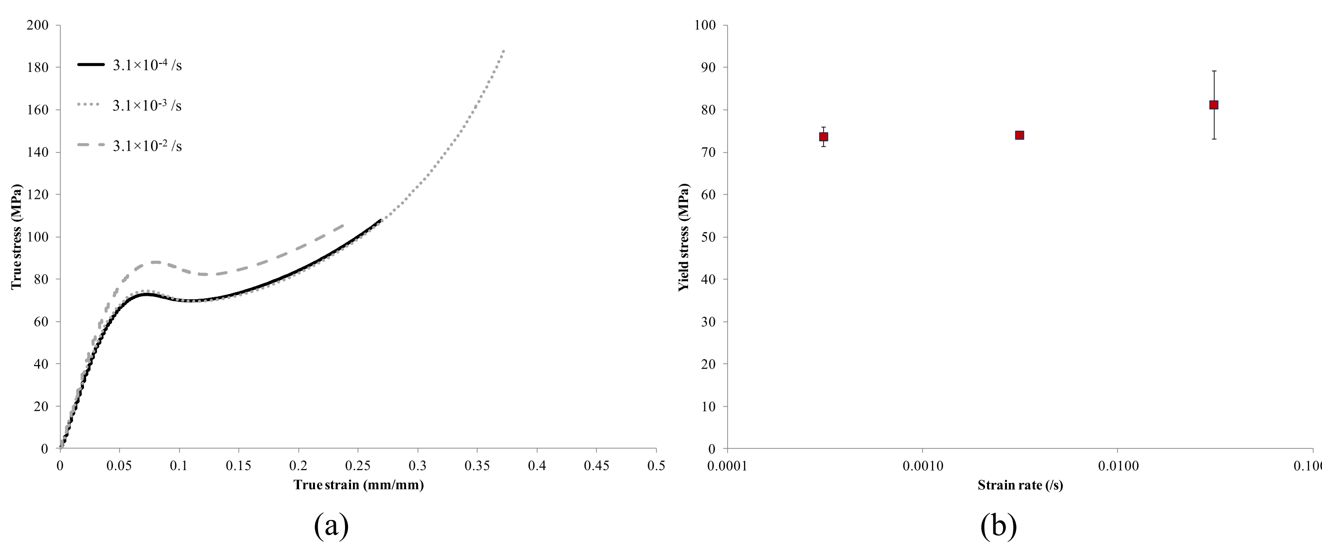
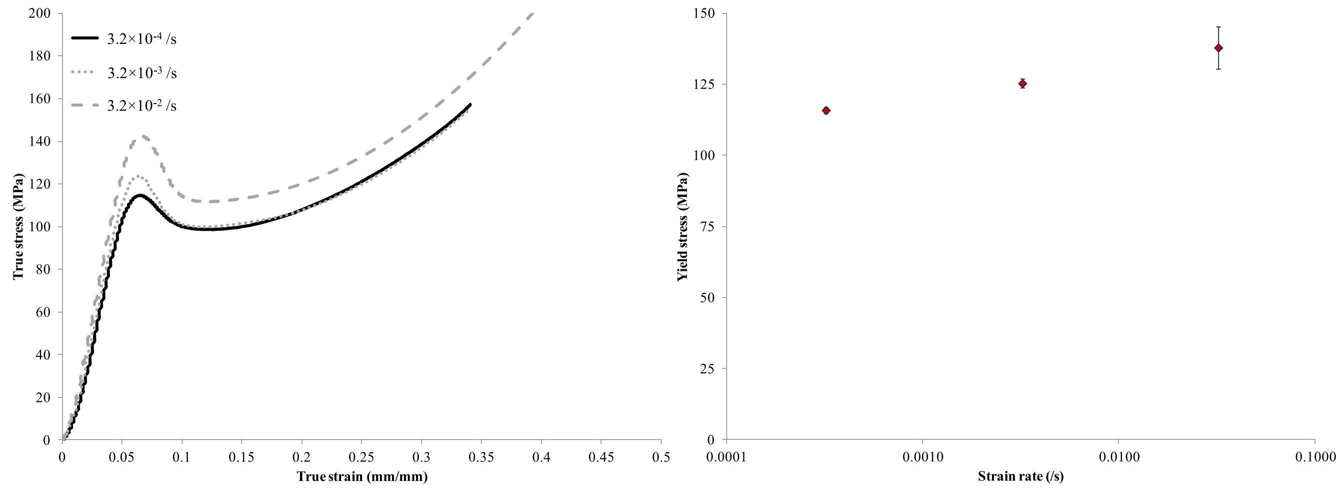


Fig. 1. PC – (a) Representative stress–strain plot in compression; (b) true compressive yield stress as a function of strain rate.



(a) (b)

Fig. 2. SAN – (a) Representative stress–strain plot in compression; (b) true compressive yield stress as a function of strain rate.

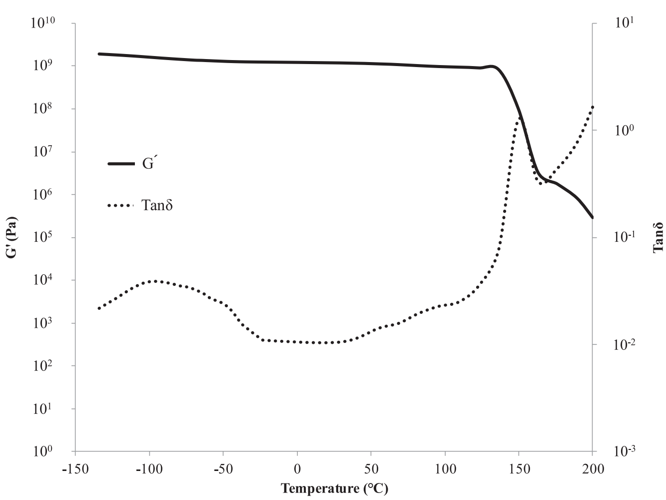


Fig. 3. PC storage modulus and loss tangent as a function of temperature at 1 Hz.

plot is decomposed into α and β component according to the DSR procedure described by Mulliken and Boyce [25]. Fig. 6 shows the plot of storage modulus separated into their respective α and β components at a particular strain rate. Analytical expression was used to decompose the storage modulus plot into α and β components. From the plot, the onset of β relaxation can be easily observed. Now, from Fig. 4, a shift in β transition of 15.1 °C/decade of strain rate is calculated. This shift is used to find the range of strain rates between which the β relaxation is activated at room temperature (296 K). By shifting the plot of onset of β relaxation according to the procedure described earlier, the predicted transition can be found as shown in Fig. 7. According to the figure, the onset of β relaxation can be predicted to occur between strain rates of 2/s and 20/s at room temperature for PC.

From the work done by Bauwens, Bauwens-Crowet and Homès [30–33,36] on Makrolon PC to describe the rate dependent behavior of amorphous polymer, the critical strain rate at room temperature is calculated to be 14.5/s, which falls within the range calculated in this study. It should be noted that the PC sample used by Bauwens-Crowet et al. [30] is Makrolon from Bayer as in this study, although the grade is not known in their case (Makrolon

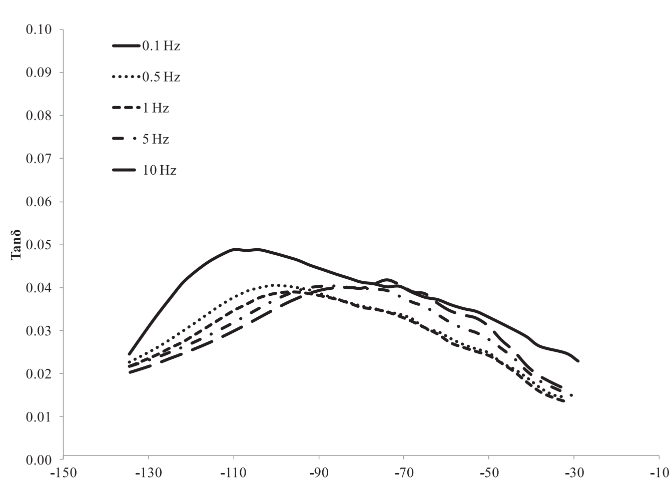


Fig. 4. Shift in β transition with increasing frequency in PC.

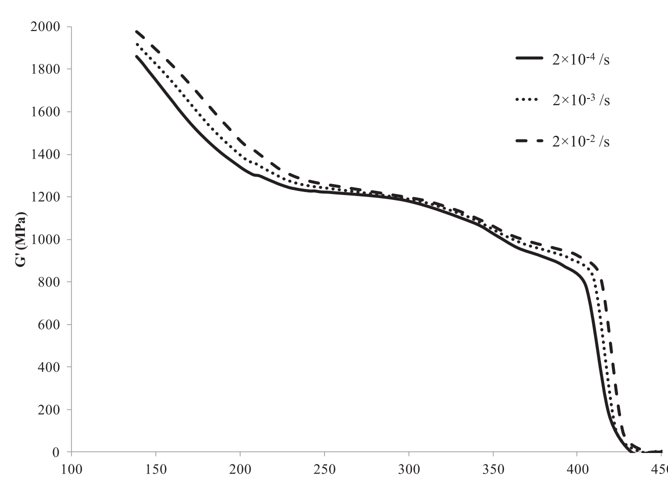


Fig. 5. Dependence of PC storage modulus on strain rate as a function of temperature (°K).

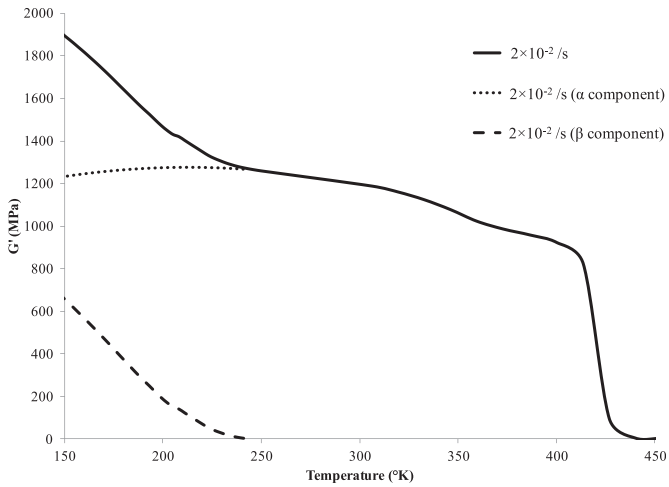


Fig. 6. Decomposition of PC storage modulus into α and β components at a strain rate of 2×10^{-2}/s.

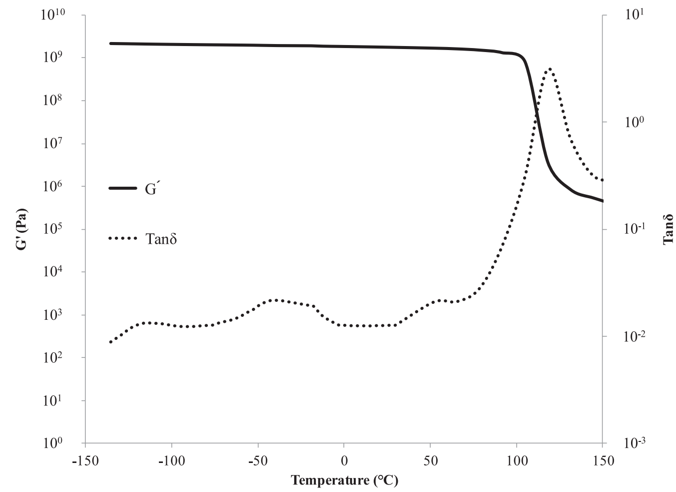


Fig. 8. SAN storage modulus and loss tangent as a function of temperature at 1 Hz.

2800 was used in this study). Even though a smooth rather than sharp transition from single activated process to multiple rate activated process occurs in polymers, the fact that the same Makrolon PC was used in this study and by Bauwens-Crowet et al. [30], the similarities in activation energies for β transition and β yield process and in the critical strain rate for onset of β relaxation in both cases suggest that the high strain rate behavior of PC can be extrapolated from the work done by Bauwens-Crowet et al. [30].

4.2.2. SAN system

Representative storage moduli and loss tangent plots in logarithmic scale obtained in DMA test for SAN are shown in Fig. 8. A broad secondary transition or a combination of multiple secondary transitions can be observed in SAN according to the loss tangent plot. Although a distinctive shift in α transition can be observed with increasing frequency or strain rate, no such shift can be observed with increase in strain rate in the broad secondary peak. Fig. 9 shows the dependence of storage modulus on strain rate as a function of temperature in SAN obtained *via* DMA test. As shown in the plot, the storage modulus decreases almost linearly with increasing temperature up to α transition at all three strain rates.

Similar storage modulus behavior has been reported in the literature in the case of PMMA [25]. It was hypothesized that the onset of β relaxation in that particular case occurs at a low strain rate of 10^{-4}/s–10^{-5}/s at room temperature. The plot of compressive yield stress as a function of strain rate revealed that indeed the β yield process for PMMA started at a low strain rate. Thus, a linear dependence of compressive yield stress on logarithm of strain rate from a low strain rate to a high strain rate can be considered [25]. A similar observation can also be made based on the storage modulus and loss tangent data obtained from DMA tests for SAN in this study. Therefore, the linear dependence of yield stress on logarithm of strain rate at low strain rates obtained by plotting the experimental data discussed in Section 4.1 can be used to predict the behavior at high strain rate in the case of SAN.

4.3. Coefficient of friction

Fig. 10 shows the COF, μ, measured for both PC and SAN. An average COF value of 0.6 for PC and 0.45 for SAN system is reported. It should be noted that, this COF values are higher than the values reported in literature [7] for the same model systems. The higher values of COF can be attributed to the heat treatment process employed in this study as it eliminates chain orientation, and, thus, produces unoriented surface compared to the oriented surface due to injection-molded specimen used in the literature. Since, the surface of both PC and SAN samples can be considered smooth

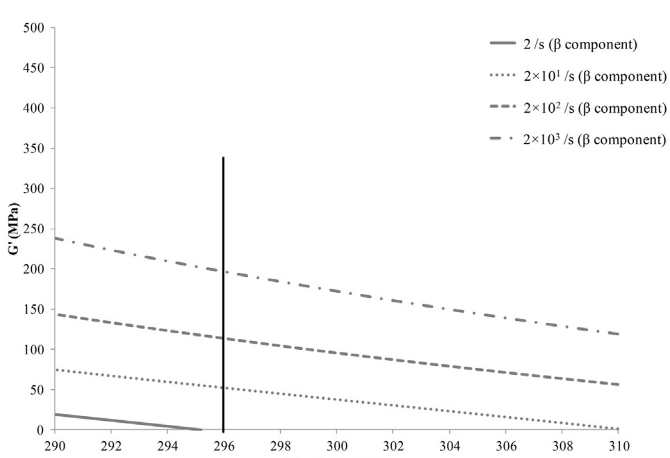


Fig. 7. Prediction of onset of β relaxation in PC. The vertical line denotes room temperature.

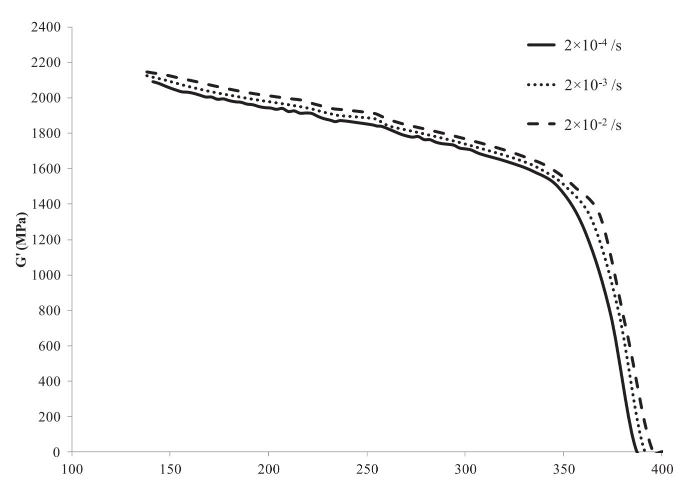


Fig. 9. Dependence of SAN storage modulus on strain rate as a function of temperature.

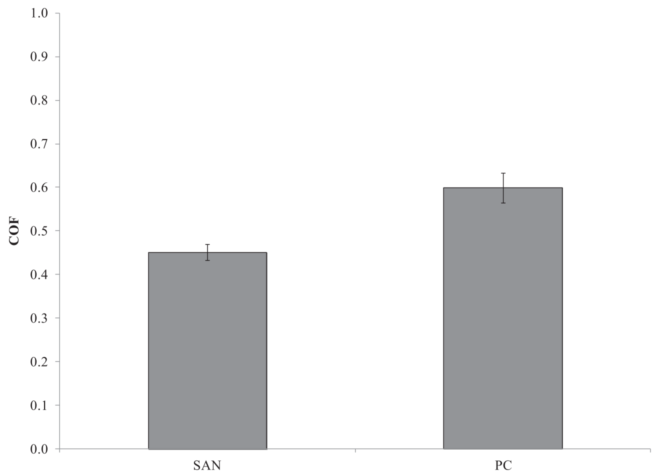


Fig. 10. Coefficient of friction measured using 10 mm × 10 mm flat tip for the model systems.

according to the roughness values reported in Section 3.1, this COF value can be used to describe the frictional behavior at a low pressure in the pressure dependent frictional model for FEM simulation, to be described later.

4.4. Viscoelastic recovery

The viscoelastic recovery or percent relaxation of PC and SAN scratch has been investigated by comparing the scratch depth and shoulder height measured right after the scratch test with that measured 8 months after the scratch test. The samples were stored in a dessicator after the scratch test and considered to be fully relaxed after this time period. The percent relaxation is calculated using the following equation:

$$\%\text{Relaxation} = \frac{SD_{residual} - SD_{relaxed}}{SD_{residual}} \times 100 \tag{8}$$

Where, $SD_{residual}$ is the residual scratch-induced deformation (scratch depth, shoulder height) measured right after the scratch test and $SD_{relaxed}$ is the scratch-induced deformation (scratch depth, shoulder height) measured after 8 months of the scratch test. Same points along the scratch path were taken for the measurement of viscoelastic recovery. For PC, more than one point at a particular location (525 μm × 700 μm window) is compared to give an average value and standard deviation. For SAN, only one point at a particular location is considered.

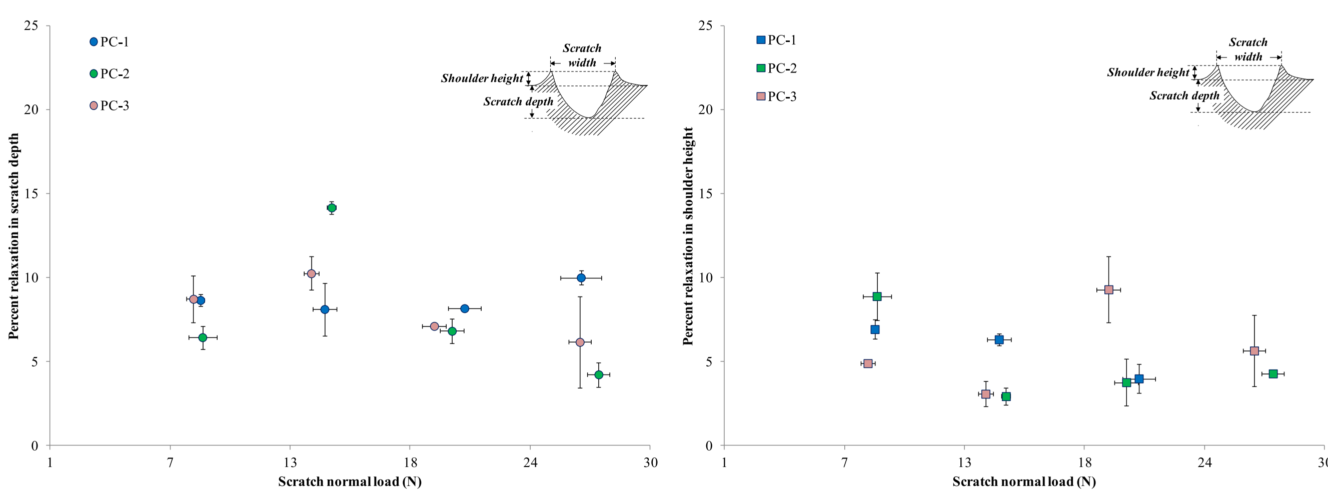
Figs. 11 and 12 show the percent relaxation in scratch depth and shoulder height for PC and SAN, respectively. As can be seen in the figures, in general, the viscoelastic recovery is less than 10% for PC and less than 7% for SAN, which can be considered minor. It is noted that the percent relaxation in shoulder height is less than that of scratch depth. Furthermore, since both shoulder height and scratch depth show positive percent relaxation, the scratch groove can be considered to become flatter with time for both PC and SAN.

According to the experimental results, the viscoelastic recovery or percent viscoelastic relaxation can be considered insignificant for both PC and SAN scratch. Thus, the viscoelastic contribution can be neglected in the constitutive model used for quantitative prediction of scratch-induced deformation *via* FEM.

5. FEM modeling

A commercial finite element package ABAQUS® [73] (V. 6.9) was employed to conduct the 3-D FEM simulation of scratch behavior of PC and SAN model polymers, using the supercomputing facilities at Texas A&M University. The study focuses on simulating the development of residual shoulder height, scratch depth and scratch width along the scratch path due to the application of linearly increasing normal load according to the ASTM standard for scratch testing [16]. It should be noted that, the development of micro-cracking, crazing and fish-scale formation in the scratch groove during the scratch process was not included in this FEM simulation. Also, as discussed earlier, since there is negligible thermal effect on scratch behavior of polymers at the scratch speed employed in the experiment and load range selected for FEM simulation, the temperature dependent constitutive behavior was not considered. Furthermore, as the viscoelastic effect on scratch behavior of SAN and PC model systems has shown to be insignificant; viscoelasticity was not considered. Since the scratch process is essentially dynamic in nature, dynamic stress analysis with explicit scheme was employed in the FEM simulation.

Drucker and Prager [74] proposed a criterion to include pressure sensitive yield behavior of materials. According to the Drucker-



(a) (b)

Fig. 11. Percent viscoelastic recovery in – (a) scratch depth; (b) shoulder height; in PC.

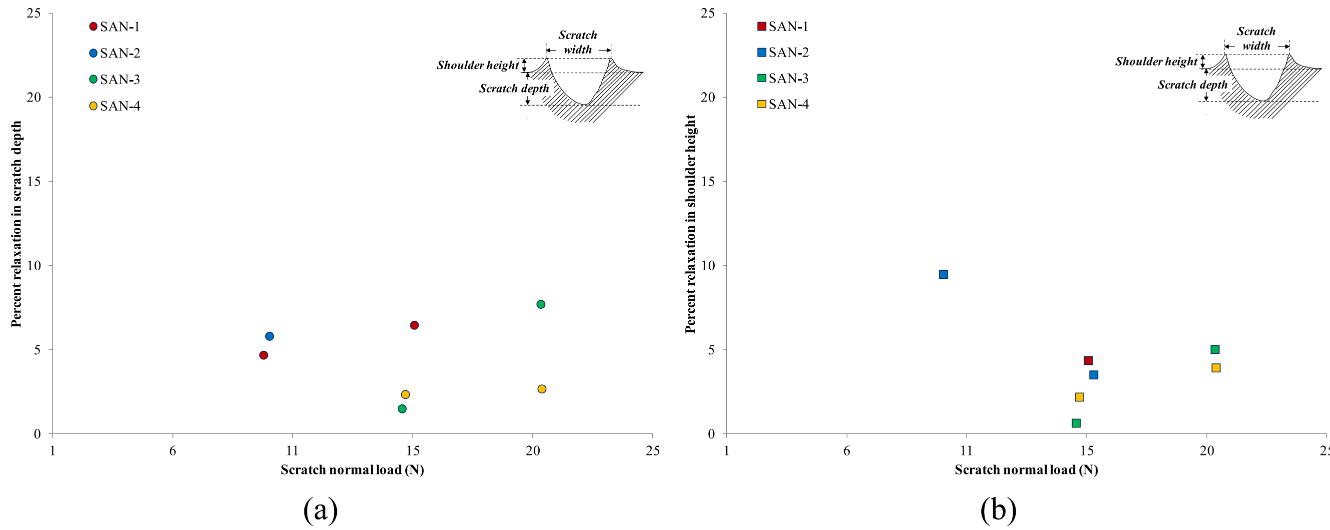


Fig. 12. Percent viscoelastic recovery in – (a) scratch depth; (b) shoulder height; in SAN.

Prager criterion, plastic yielding begins when the J_2 invariant of the deviatoric stress tensor and the hydrostatic pressure, P, reach a critical combination [75]:

$$\sqrt{J_2(s)} + \eta P = c \tag{9}$$

Where, η and c are material parameters. The von Mises yield locus is recovered from the Drucker-Prager criterion when $\eta = 0$. As discussed earlier, since polymers in general show linear dependency on hydrostatic pressure, linear Drucker-Prager model [73], a built-in inelastic material model in ABAQUS®, was employed to describe the rate and pressure sensitive constitutive relationship of the PC and SAN model systems. The true yield stress and post-yield behavior in uniaxial compression at different strain rates was used for that purpose. The yield criterion can be written as [73]:

$$F = t - P\tan\delta - \left(1 - \frac{1}{3}\tan\delta\right)\sigma_c = 0 \tag{10}$$

Where, $t = \frac{1}{2}q\left[1 + 1/K - (1 - 1/K)(r/q)^3\right]$, P is the hydrostatic stress $\left(P = -\frac{1}{3}\mathrm{trace}(\boldsymbol{\sigma})\right)$, δ is the pressure sensitivity index, σ_c is the true yield stress under uniaxial compression, K is the ratio of the yield stress in triaxial tension to the yield stress in triaxial compression, q is the von Mises equivalent stress $\left(q = \sqrt{\frac{3}{2}(\boldsymbol{S}:\boldsymbol{S})}\right)$, $\boldsymbol{S}$ is the deviatoric stress ($\boldsymbol{S} = \boldsymbol{\sigma} + P\boldsymbol{I}$), and r is the third invariant of deviatoric stress $\left(r = \left(\frac{9}{2}\boldsymbol{S}\cdot\boldsymbol{S}:\boldsymbol{S}\right)^{\frac{1}{3}}\right)$. The flow potential is chosen as: $G = t - P\tan\psi$, where ψ is the dilation angle. If $\psi = 0$, the plastic deformation is incompressible; if $\psi \geq 0$, the material dilates.

For yield stress of PC, as discussed earlier, at low strain rate, experimental data obtained in uniaxial compression test was provided. To provide medium to high strain rate behavior in uniaxial compression, extrapolated data from Bauwens-Crowet et al. [30] was incorporated as shown in Fig. 13(a). The post-yield behavior of PC in uniaxial compression, reported in the literature and also observed in this study, showed to be insensitive to strain rate. Thus, the post-yield behavior at different strain rates were given by increasing the stress and strain similar to the increment in yield stress and yield strain with strain rates, as shown in Fig. 13(b). For SAN, as discussed earlier, the high strain rate behavior was predicted using the linear dependency of compressive yield stress on logarithm of strain rate obtained experimentally for low strain rates (Fig. 14(a)). The post-yield behavior of SAN at different strain rates

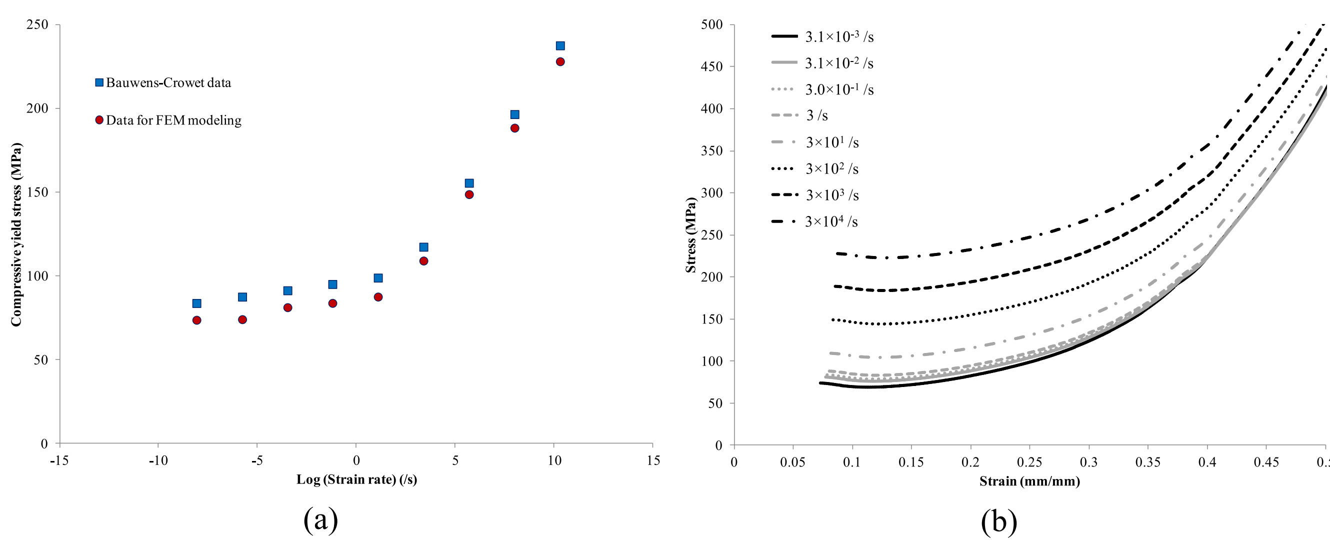


Fig. 13. (a) Yield stress and (b) post-yield behavior of PC in uniaxial compression used for FEM modeling.

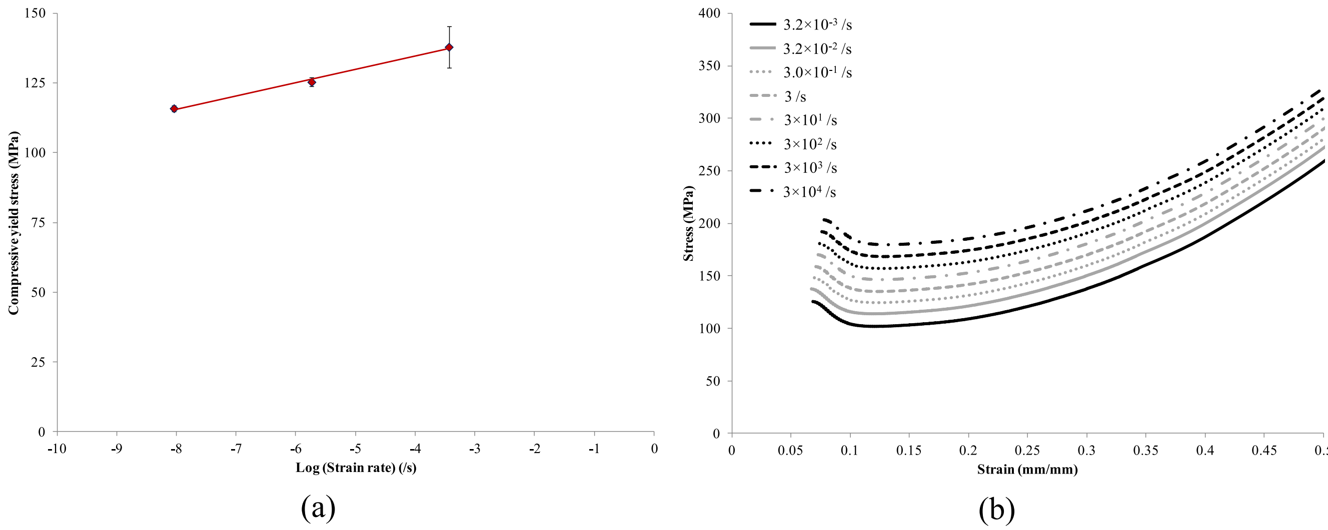


Fig. 14. (a) Yield stress and (b) post-yield behavior of SAN in uniaxial compression used for FEM modeling.

was given following the similar procedure described for PC as shown in Fig. 14(b). In both PC and SAN, yield strain was assumed to follow the experimental data obtained at lower strain rates.

To calculate δ, uniaxial tension test data is required. For tensile yield stress of PC, experimental data obtained in uniaxial tension test at low strain rates was used, and, for medium to high strain rate, extrapolation using data from Bauwens-Crowet et al. [30] was used. The yield stress values in uniaxial compression and tension were used to calculate δ. Using the value of δ, K was calculated using an option provided in ABAQUS® to match Mohr-Coulomb and linear Drucker-Prager model parameters for materials with low frictional angle so that both models provide the same failure definition in triaxial compression and tension. The dilation angle, ψ was assumed to be zero since, as discussed earlier, polymers have shown insignificant plastic dilation.

Since $\delta \neq \psi$, non-associative flow is considered in this study. Now, since δ and K are obtained from yield stress values in tension and compression, which are rate dependent, these values also vary slightly with strain rate. Since the rate dependent true stress–strain relationship in uniaxial compression were the primary input in describing the constitutive behavior and it has been shown that [5] tensile behavior has little influence on scratch depth and shoulder height formation during scratch, these minor variations can be considered trivial. Thus, the values of δ and K were provided at an average strain rate (~480/s for PC, ~550/s for SAN) which the polymer surface would experience using the scratch width calculated from Hertzian analysis [20] in Equation (1). For SAN, the calculation is more complicated since it shows brittle behavior in tension and the tensile strength remains almost unchanged with increasing strain rate. Thus, for SAN, the tensile strength value was used along with compressive yield stress to calculate δ and K. Table 2 lists the values of the parameters of linear Drucker-Prager model used in the FEM simulation.

Table 2
Values of linear Drucker-Prager model used in the FEM simulation.

Parameter	PC	SAN
δ	27.13°	40.24°
K	0.85	0.78
ψ	0	0

Linear elasticity before yielding is considered to describe the elastic regimes of PC and SAN model systems. As a result, the secant modulus, calculated based on the yield stress and yield strain in uniaxial compression, is considered to be the same as the Young's modulus. Since the difference in tensile and compressive modulus was observed to be minimal, uniaxial tensile and compressive moduli were considered to be the same at all strain rates. Also, as has been pointed out in the literature [3,76] that the modulus in the range of 1.65 GPa–4 GPa has negligible effect on scratch depth for the scratch tip geometry employed in the study, which is the case for both PC and SAN, it is reasonable to treat the secant modulus of PC and SAN as rate independent. Therefore, the modulus values at an average strain rate of ~480/s for PC and ~550/s for SAN were used for the modeling. Table 3 lists the secant modulus and Poisson's ratio values used for the FEM modeling.

To define the contact between the rigid spherical tip and polymer substrate during the scratching process, finite sliding contact pair algorithm in ABAQUS® was used. Pure master-slave contact algorithm was employed where the tip is considered to be rigid and the polymer substrate is considered to be deformable. To define the frictional behavior of the interacting surfaces, isotropic Coulomb friction model provided by ABAQUS® was used. This friction model allows defining friction coefficient as a function of contact pressure. The coefficient of adhesive friction was included in the FEM model based on Equation (5). As discussed earlier, the shear yield stress in sliding contact can be considered independent of strain rate, and, thus, the shear yield stress at low strain rate can be used to calculate coefficient of friction due to adhesion. To calculate the shear yield stress, the pressure dependent yielding plot constructed based on the uniaxial tension and compression data at a strain rate of ~3.1 × 10^{-3}/s was used. A pressure dependent shear yield stress plot was generated using the relation: $\tau_y = \sigma_y/\sqrt{3}$. Since this plot is essentially shear yield stress *vs.* hydrostatic pressure and the input in friction model is contact pressure dependent, the slope of this pressure dependent shear stress plot was divided by 3 and is taken as the pressure coefficient, γ. The shear stress value when hydrostatic pressure is zero is taken as τ_0. Although, SAN shows brittle behavior in tension and Equation (5) is generally applicable for ductile polymers, similar procedure was applied to obtain

Table 3
Secant modulus and Poisson's ratio values used in the FEM simulation.

	PC	SAN
Secant modulus, GPa	1.86	2.45
Poisson's ratio	0.37	0.35

Table 4
Values used for calculating coefficient of adhesive friction using Equation (5).

	PC	SAN
τ_0, MPa	38.32	51.73
γ	0.06	0.16

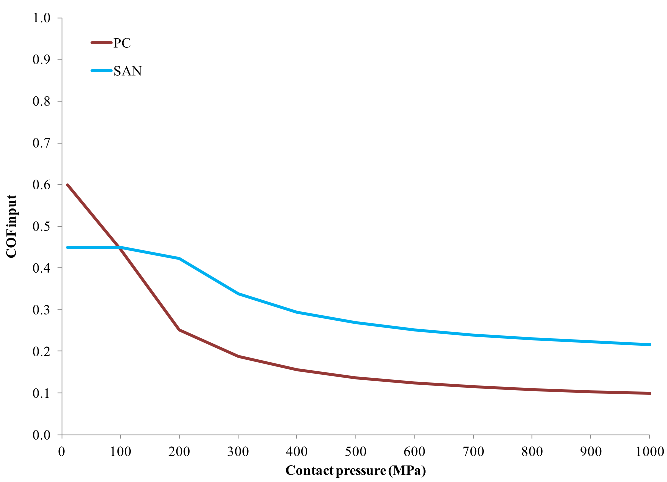


Fig. 15. Friction model used in the FEM simulation.

coefficient of adhesive friction data to include in the FEM model for SAN. Table 4 lists the values of the parameters calculated from pressure dependent uniaxial data. Using the pressure dependent coefficient of adhesive friction values obtained from Equation (5) and Table 4, the friction model for FEM simulation was described. Fig. 15 shows the pressure dependent friction model used in the simulation of PC and SAN. The coefficient of friction values obtained experimentally for PC and SAN using the flat stainless steel tip (Section 4.3) were used as an upper limit for coefficient of friction due to adhesion in the friction model by assuming that the values obtained were at low pressure.

The FEM computational domain used to perform numerical analysis is shown in Fig. 16. A mesh with 512 elements along the critical length (A–B) was chosen in this study rendering an element dimension of 22.8 μm × 28.1 μm × 33.3 μm. This particular meshing was chosen based on the investigation on effect of meshing on scratch-induced deformation to find an optimum meshing that would provide sufficient accuracy with less computation time required; can be found elsewhere [47]. The model geometry and boundary conditions applied in this study were the same as described in the literature [5]. The FEM simulation of scratch-induced deformation was divided into three steps; indentation, scratching and tip removal; following the ASTM/ISO standard for scratch testing [16], described elsewhere [5]. The difference between instantaneous scratch depth and the amount of scratch depth recovered due to elastic recovery is denoted as scratch depth (residual scratch depth) in this work. As the half model was used in this simulation due to symmetry condition, the normal load applied on the scratch tip during the simulation was half the actual value of the normal load. To save computation time of the numerical simulation, the tip moves over a length of 12 mm at a constant scratch speed of 10 m/s was specified in the numerical simulation. The rate dependent constitutive relationship was scaled accordingly (e.g., data corresponding to a strain rate of 3/s was given as 300/s) to simulate the scratch behavior at 100 mm/s scratch speed. The validity of the usage of this method was checked and confirmed, and, is presented elsewhere [47].

6. Experimental validation of fem simulation

6.1. PC system

Figs. 17–19 show the comparison of FEM simulation and experiment on residual scratch depth, shoulder height and scratch width formed during the scratching process in PC, respectively. PC-1 to PC-5 denote the experimental measurements of the five scratches on the sample. As can be seen in the figures, all five scratches show similar scratch-induced deformation highlighting the repeatability and consistency of the scratch testing methodology. Although, the scratch testing was experimentally carried out up to a scratch normal load of 70 N, the deformation analysis was done in 1–35 N load range since this was the load range quantitative prediction of scratch-induced deformation using FEM was performed.

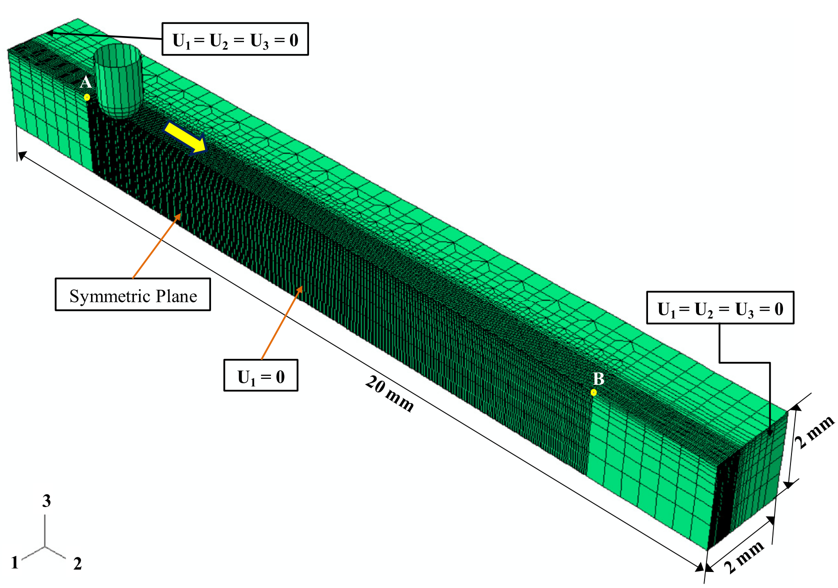


Fig. 16. FEM model showing mesh and boundary conditions employed.

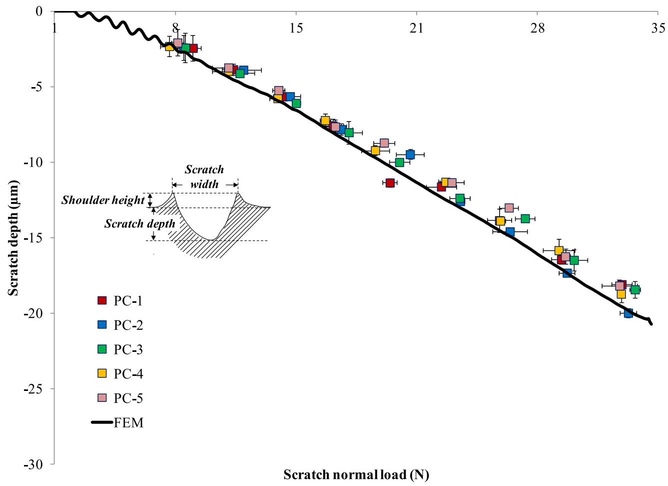


Fig. 17. Comparison of FEM simulation and experiments on scratch depth evolution as a function of scratch normal load for PC.

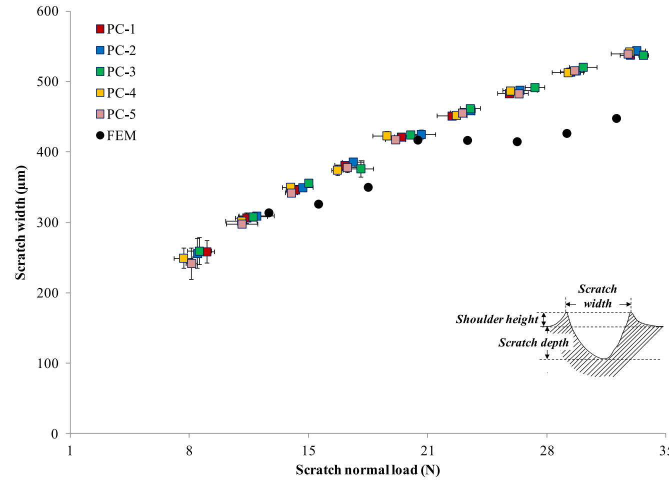


Fig. 19. Comparison of FEM simulation and experiments on development of scratch width as a function of scratch normal load for PC.

As shown in the figures, the scratch depth prediction using FEM matches well with the experiment. For shoulder height and scratch width, the FEM simulation predicts well until around 21 N. Beyond that, the FEM under-predicts the shoulder height and scratch width formation. It has been shown that at higher loads the contribution of scratch groove formation comes not only from the elastic recovery but also from the material displacement from the front of scratch tip towards the side [77]. This displaced material added on to the shoulder height formed due to elastic recovery, which may be the primary reason for discrepancy between FEM simulation and experiment in shoulder height and scratch width at higher loads. The topographical images of the side of scratch groove obtained *via* VLSCM for PC showed increase in surface roughness with increasing normal load, which also corroborates the fact that material in front of scratch tip was indeed moved towards the side. Since, the Lagrangian analysis used in this FEM simulation is unable to simulate the bulk material movement, the discrepancy at higher normal load in case of shoulder height and scratch width is expected. Nevertheless, FEM simulation of scratch behavior of PC shows reasonable success in predicting the scratch-induced deformation up to plowing stage, where material removal becomes dominant.

Fig. 20 shows the comparison of scratch coefficient of friction (SCOF) obtained during experiment (SCOF $= F_t/F_n$) for PC with the friction value obtained from FEM simulation by adding the coefficient of adhesive friction and coefficient of friction due to deformation using Equation (2). The coefficient of friction due to deformation is calculated using both Equations (6) and (7). The contact radius, a, required for the equations is calculated using the following relation:

$$a = \sqrt{2dR - d^2} \tag{11}$$

Where, R is the radius of the scratch tip and d is the instantaneous scratch depth obtained *via* FEM simulation. The fluctuations at the beginning of the experimentally obtained SCOF data are due to the inertia effect of the scratch tip movement. When the scratch tip starts moving from static position to 100 mm/s, it has to accelerate within short time which causes the fluctuation in the initial portion of the experimental data.

As pointed out earlier, this method of dividing the friction contribution is generally applicable for ductile polymers, which is the case. As shown in the figure, the FEM simulation matches quite well with the experiment. It should be noted that, as discussed

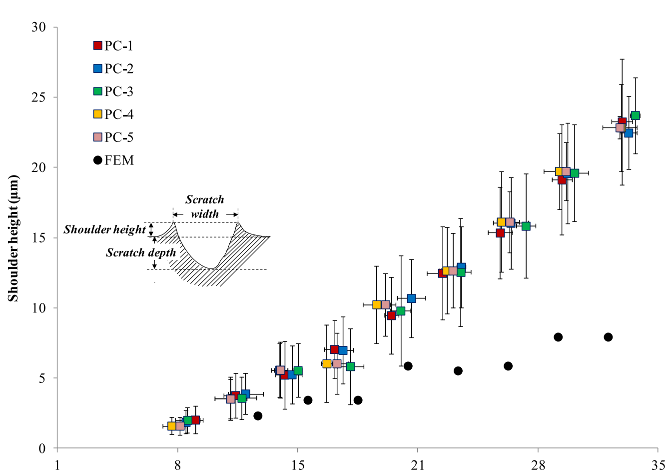


Fig. 18. Comparison of FEM simulation and experiments on development of shoulder height as a function of scratch normal load for PC.

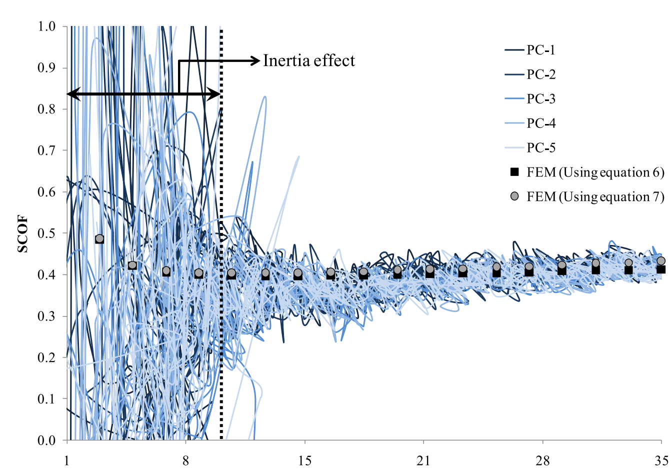


Fig. 20. Comparison of scratch coefficient of friction (SCOF) obtained *via* experiments with development of friction calculated using FEM simulation for PC.

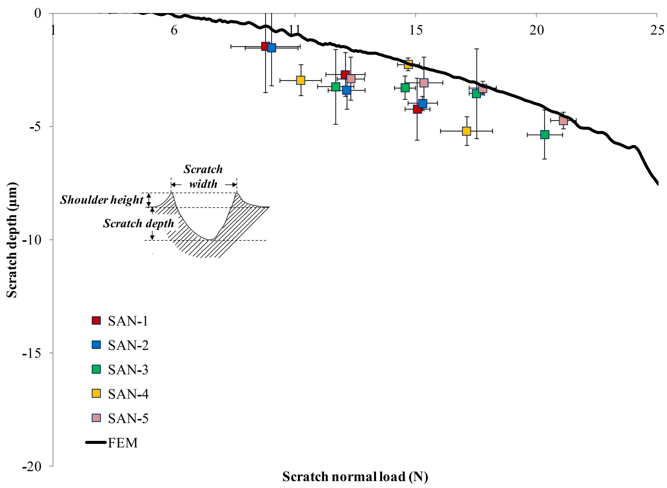


Fig. 21. Comparison of FEM simulation and experiments on scratch depth evolution as a function of scratch normal load for SAN.

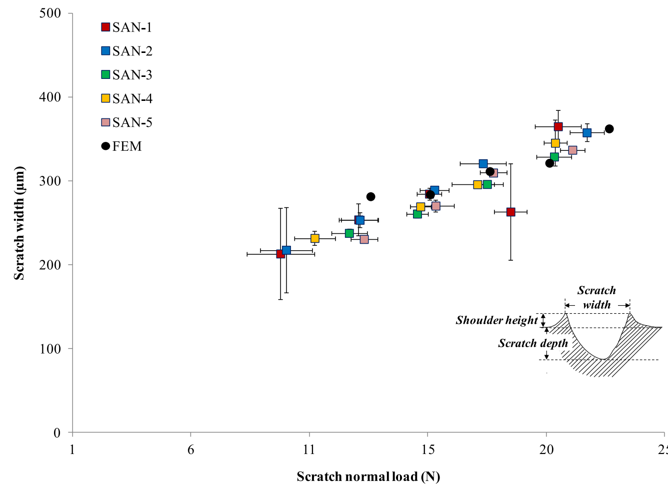


Fig. 23. Comparison of FEM simulation and experiments on development of scratch width as a function of scratch normal load for SAN.

earlier, the calculation of COF due to deformation is simplistic and also there might be an interaction between adhesion and deformation part of friction which the hypothesis did not take into account. Nevertheless, by employing the simplistic method of calculating frictional value using Equation (2), FEM simulation predicts the SCOF quite well.

6.2. SAN system

Figs. 21–23 show the comparison of FEM simulation and experiment on residual scratch depth, shoulder height and scratch width formed during the scratch process in SAN, respectively. SAN-1 to SAN-5 denote the experimental measurements of the five scratches on the sample. Again, the repeatability and consistency of the scratch testing procedure is self-evident from the figures. For SAN, micro-cracks/crazes formed in the scratch groove along the scratch path at higher loads. Thus, measurement of scratch-induced deformation was done only up to the point of onset of micro-cracking/crazing for all the scratches and plotted accordingly. As shown in the figures, the scratch depth and scratch width prediction using FEM matches well with the experiment. For shoulder height, similar reasoning as discussed earlier based on a previous study [77] can be applied to explain the discrepancy in FEM simulation and experiment for SAN. In addition, the fact that the method used in this study to calculate COF due to adhesion is generally only applicable for ductile materials and SAN shows brittle behavior in tension may also contributes to the discrepancy.

Similar to PC, Fig. 24 shows the comparison of SCOF obtained during experiment for SAN with the friction value obtained from FEM simulation using Equation (2). Again, the fluctuations at the beginning of the experimentally obtained SCOF data for SAN are due to the inertia effect of the scratch tip movement. As shown in the figure, the FEM simulation over-estimates the scratch coefficient of friction. Since SAN is brittle in tension, this simulation results indicate that the frictional calculations used in this study, which is essentially for ductile polymers, can be of limited validity for brittle polymers.

It should be noted that a deeper scratch depth and higher shoulder height for both PC and SAN model systems is observed in this study compared to the same systems investigated in the literature [7]. The literature study did not carry out heat treatment on the samples. Since the injection-molded specimen has preferred chain orientation along the mold direction, a lower value of COF is expected along the chain orientation compared to the unoriented

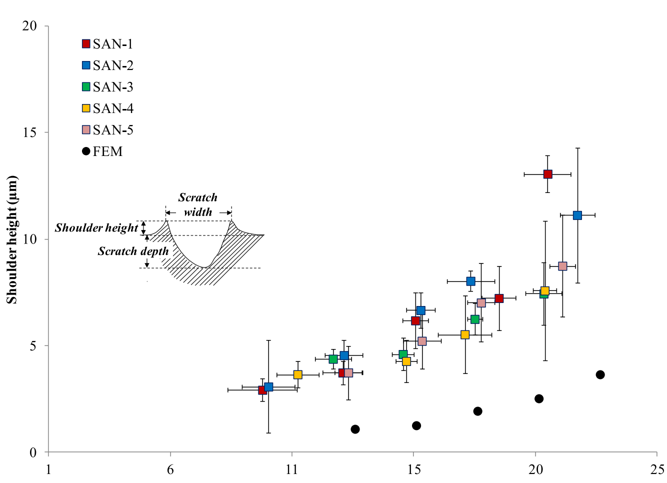


Fig. 22. Comparison of FEM simulation and experiments on shoulder height evolution as a function of scratch normal load for SAN.

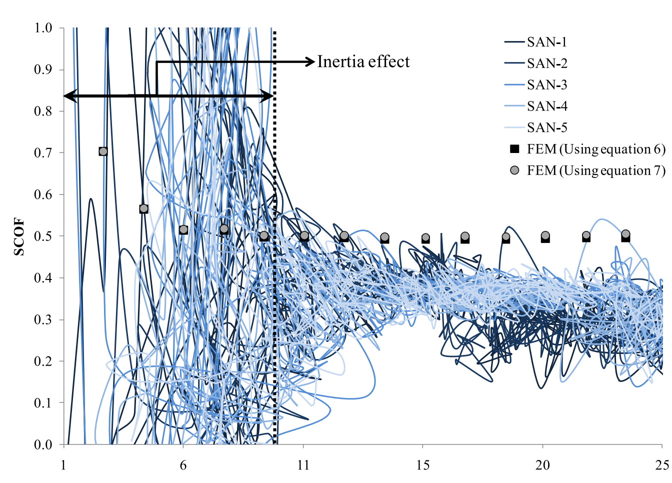


Fig. 24. Comparison of scratch coefficient of friction (SCOF) obtained *via* experiments with development of friction calculated using FEM simulation for SAN.

structure due to heat treatment [59,60] which contributes to shallower scratch depth and lower shoulder height reported in the literature for the same systems.

The discrepancy in shoulder height between FEM simulation and experimental observation for both model polymers can be attributed to the bulk material flow from the scratch front towards the sides at higher loads, which contribute to the shoulder height pile-up. In this study, the ABAQUS® default Lagrangian analysis was used which did not take into account the bulk material flow. The discrepancy in FEM simulation and experiment can also be partially attributed to the simplistic constitutive model employed in this study. Appropriate contact mechanics model is needed for brittle polymers. Therefore, an appropriate constitutive relationship taking into account the aforementioned factors with adequate frictional model is expected to provide more accurate prediction of scratch-induced deformation in polymers. Nevertheless, with the simplistic material and frictional model employed here, the FEM simulation is able to correlate well with the scratch-induced deformation of PC and SAN model polymers. It is expected that the same framework described in this study can be utilized for predicting scratch-induced deformation of other amorphous polymers. The usefulness of the present research for selection and design of scratch resistant amorphous polymers is anticipated. It should be noted that for semi-crystalline polymers, the crystallinity, crystal orientation, skin-core morphology, and the frictional characteristics, just to name a few, should also be considered when modeling using the framework described in this study. These factors are by no means easy to address. They are subjected to our future investigation.

7. Conclusions

Using the Ree-Eyring theory and experimental data at lower strain rates, the mechanical behavior of model polymers at high strain rate was extrapolated for modeling scratch behavior at high strain rates. A pressure dependent frictional model was developed and included in the FEM model. The experimental study on model polymers shows that viscoelasticity of polymers plays a minor role on the scratch behavior. Good agreement between the experimentally observed scratch-induced deformation and the FEM model is obtained. Therefore, the choices of the constitutive relation, frictional behavior, and assumptions included in the FEM modeling are considered appropriate. Thus, by knowing the constitutive behavior and surface frictional characteristics, one can predict the scratch-induced deformation of polymers using the modeling scheme carried out in this study. Consequently, the present research is expected to facilitate fundamental understanding and designing of scratch resistant polymers for a vast variety of engineering applications.

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